

Dice Palette:VR Authentication based on Freehand 3D Interaction

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ABSTRACT

At present, most VR authentication studies directly transplant 2D mobile authentication mechanisms into 3D environments, which not only ignore the efficiency of such authentication mechanisms in virtual environments, but also may cause security problems since surrounding bystanders can easily guess the password from users' gestures. Considering these problems, we develop an authentication system called Dice Palette, which supports 3D freehand interaction in user registration and login. Compared with previous methods, Dice Palette improves security by avoiding bystanders from guessing gestures. On the other hand, it significantly enhances password capacity, effectiveness and immersion of the interaction.

Index Terms: Human-centered computing—Interaction design—Interaction design process and methods—; Security and privacy—Domain-specific security and privacy architectures—

1 INTRODUCTION

Unique immersion and natural interactive experience have always been the focus of continuous exploration and breakthrough in various fields of virtual reality. Although there have been extensive studies on VR authentication, the majority of them still use a 2D authentication interface. Numerous studies on hand interaction-based VR authentication have been conducted. For example, a gesture unlocking 2D or 3D patterns in space can be stored as authentication password [4, 5]. Such authentication design generally has a simple layout of patterns or passwords in the space, and it is easy for bystanders to directly observe and remember users' gestures, which induces security problems as shown in Figure 1(a). Moreover, most studies directly transplant traditional authentication mechanisms (digital PIN, unlock pattern etc.) into VR systems [1, 4, 5], which lead to applicability and security issues. According to studies conducted by George et al. on the issue of translating 2D unlocking methods directly to VR, 18% of users' digital PINs and unlock patterns might be obtained by watching users gestures [2].

Some researches have made corresponding designs according to the characteristics of VR and gesture interaction [3]. However, in terms of immersion and natural interaction, these investigations do not provide new insights concerning VR authentication. VR authentication can be designed in multiple and innovative ways that are more suitable for the VR environment, while maintaining its convenience, effectiveness and security. In this paper, we introduce Dice Palette, which is a VR password authentication system based on freehand interaction. A combination of numbers one to six and five colors (red, green, blue, yellow and black) is used in password design. Due to VR's unique immersive characteristics and diverse forms of interaction, we hope to design a VR authentication system in a more interesting, novel and entertaining way, rather than simply extending traditional methods, such as directly transplanting graphical pattern



Figure 1: (a) A bystander deciphers a user's password by his gestures. (b) A screenshot of user interactions in Dice Palette.

and digital PIN unlock to VR for redesign. The design of Dice Palette was inspired by common objects in life, combined with real freehand interactions, allowing users to quickly understand and experience the whole VR authentication with natural immersion and presence.

2 DESIGN METHOD

There are three primary components of Dice Palette as shown in Figure 1(b).

(1) Dice: A 6-sided dice consisting of numbers 1-6 near the right hand. The dice volume is close to the size of the virtual hand in the grasp position;

(2) Palette: Palette displays five different colors (red, yellow, blue, green, and black) when users put left palms toward their faces. The palette is semi-wrapped around the right side of the left hand and is used to change the color of the numbers of the dice;

(3) Tray: Tray is settled in the front of the virtual view. It has two circular buttons and a square UI with four places (the blue button on the left is registration, and the green button on the right is login). It is used for passwords setting and validation.

2.1 Interaction procedure

The entire procedure for users to perform freehand authentication using the Dice Palette can be described as *Click-Pick-Touch-Rotate-Drop* as shown in Figure 2.

Click: Open both hands palm up on the leap motion once the user has entered the Dice Palette. The user can see two virtual hands as shown in Figure 2(1). Entering the interface for the first time, a user needs to register and set a password by pressing the blue round registration button with right hand and the Tray will turn blue. If the user has registered, click the green round button to start login as shown in Figure 2(2).

Pick up: The user uses the right hand to pick up the white dice, with numbers 1 to 6 on the six sides of the dice as shown in Figure 2(3).

Touch: To change the dice's color, the user picks up the dice and touches a color on the left palette as shown in Figure 2(4).

Rotate: The user selects a number by rotating the dice as shown in Figure 2(5).

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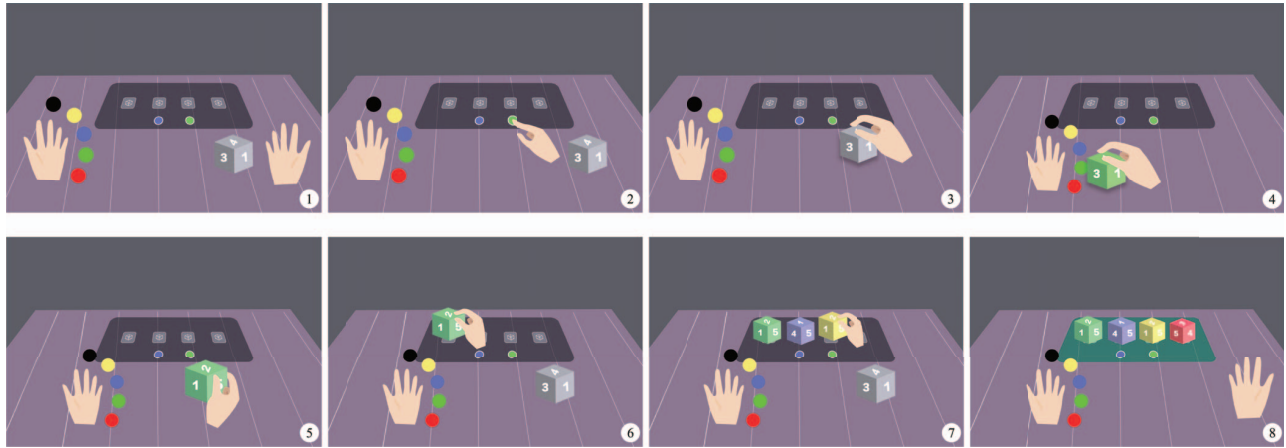


Figure 2: (1)-(8) show the *Click-Pick-Touch-Rotate-Drop* interaction procedure when the login is correct.

Drop: The user places the rotated dice on the first place of the Tray to complete the setting of the password as shown in Figure 2(6).

Once the previous place has been filled, the subsequent Dice will display and users can place all four dices in order as shown in Figure 2(7). When user presses the green round login button on the Tray after all passwords are registered and the Tray panel changes to its original color, user can use the above procedure to complete the login. For login, after all passwords are placed, the Tray turns green if password authentication succeeds (see Figure 2(8)) and turns red if authentication fails.

2.2 Password capacity

A password is verified based on the matching of both correct numbers and colors of dices. Each dice has $6 \text{ numbers} \times 5 \text{ colors}$ choices and the password is composed with four dices, so the password capacity is $30^4 = 810000$. Namely there are up to 810000 password composition ways, which effectively ensure the security of authentication compared with 10000 password capacity of 4-digit PIN.

2.3 Implementation

The Dice Palette is developed with Unity3D engine. Steam VR plug-in supports HTC VIVE head-mounted display as a virtual environment display device. Ultraleap Unity plug-in and Leap motion gesture recognition hardware are used for freehand interaction. We also add some details to the implementation of Dice Palette in terms of visual and interaction design.

Dice: Although password recognition requires a four-digit password, only one Dice is displayed when the user enters the screen. On the one hand, the presence of multiple Dice at the same time may affect the interaction experience by mistake, and on the other hand, the presence of only a single Dice can ensure the simplicity of the environment. The next Dice appears as soon as the user successfully sets the first password, giving the user feedback of “successfully set the previous password”.

Palette: Each time when the user reopens the Dice Palette, five colors randomly change their positions. This increases the security of VR authentication by preventing observers from inferring the password color from the user’s gestures.

Tray: To improve user interaction, we’ve added some interactive visual feedback details. The first is visual feedback of interacting with buttons on the tray. As the user’s hand gets close to either button, the button pops up and changes its color to distinguish it from the other button. When the round blue registration button is pressed, the Tray panel changes from black to blue, prompting the user to successfully press the button and begin registration. When

the user picks up the Dice and approaches the tray, the icon on the corresponding place will increase to indicate the spatial position of the Dice.

We show screen records of login success and failure using Dice Palette in the supplementary video.

3 CONCLUSION

In this paper, the freehand interaction capabilities of VR authentication are discussed, and a novel approach to create passwords is proposed. The authentication system need to verify both four digits and their colors to the stored password. This technique guarantees that the password is simple and straightforward, speeds up password authentication, and increases the security of password authentication. In addition, this work also selects the attributes of familiar objects such as dice and color palette as UI to change numbers and colors, which conforms to the interactive cognition and functional cognition of users when they interact by hands. We hope that our work will stimulate fresh design concepts for VR authentication in the future.

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